Conventions (apply to the whole of formB_ws.tex)

Normalization — the derivation is fully dimensionless; the bar marks the normalized version of a quantity that has a physical twin. Lengths by the particle radius R: $\bar{w} = w/R$ is the probe-center height on the wall-normal axis (the same axis the companion documents call $\bar{h}$), $\bar{w}_s$ is the contact (surface) position, nominal 1; the unknown is the offset $\bar{w}_s - 1$. Gain by $a_o = \Delta t/(\gamma_N R)$: $\bar{a} = a/a_o \in [0, 1]$ (exactly the quantity of the companion note, saturating at 1); the unbarred $a_w = a_o\, \bar{a}_w$ is the physical gain, 1/pN. Forces by a_o: $\bar{f} = a_o f$ is the normalized far-field one-step displacement of the force f, so $\bar{a}\, \bar{f} = a\, f$ identically and every equation of the derivation is dimensionless. The parameters b (a length in units of R, nominal 9/8) and p, and the noise symbols ε_w, n_w, n_a, follow the companion documents' unbarred writing; their normalization is declared here once (n_w: position noise in units of R; n_a: readout noise on the *normalized* gain, i.e. the house doc's physical readout noise divided by a_o). Background symbols: $\gamma_N = 6\pi\eta R$ (far-field Stokes drag); $\gamma(\bar{w}) = \gamma_N\, c(\bar{w})$ with $c(\bar{w})$ the wall-effect correction factor, unknown at run time — it appears only inside the $\bar{a}_w$ identity in S8, never in a run-time quantity; $\delta\bar{w}_r$ = closed-loop regulation residual (the house doc's $\delta\bar{h}_r$ renamed). Estimates wear the hat over the barred symbol: $\hat{\bar{a}}_w$, $\hat{b}_w$, $\hat{p}_w$, $\hat{\bar{w}}_s$. a_o appears only at the two interfaces: force conversion $\bar{f} = a_o f$ and the gain readout $\bar{a}_{wm} = a_{wm}/a_o$. Discrete index k, sample time Δt. (Correction log: an earlier draft wrote "$\bar{a}_w$ in 1/pN" — wrong, fixed 2026-07-31; a second draft defined e_{a_w} on the physical gain — superseded 2026-07-31 by the fully normalized scheme below. Both swaps are value-preserving: $\bar{a}\bar{f} = af$.)

Error convention. $e_\theta = \theta - \hat{\theta}$ (true minus estimate) for every estimated quantity. The gain error is $e_{\bar{a}_w} = \bar{a}_w - \hat{\bar{a}}_w$, dimensionless; with F_{dw} built from normalized efforts $\bar{f}_{dw}$, the product $F_{dw}\, e_{\bar{a}_w}$ is a normalized length, consistent with the $\delta\bar{w}$ recursion. This matches the companion note exactly: its $\bar{a}$ saturates at 1 and multiplies F_{dw} directly, which is dimensionally consistent only under the normalized-force reading.

Notation rule (user, 2026-07-31): only symbols already defined in the two companion documents are used — "Estimation of motion gain_R" and `5state_expgain_hd`; every addition needs prior discussion. Expressions are kept fully expanded — no substitution variables (house rule of `aprime_jacobian_AB.tex`).

Cross-document symbol table.

symbol	meaning in formB_ws	clash to avoid
b	Form B length scale, units of R, nominal 9/8	exponent b of `5state_expgain_hd`
p	Form B exponent, nominal 1	power p of `5state_powerlaw_hd`
$\bar{w}$, $\bar{w}_s$	height / contact position	$\bar{w} \equiv \bar{h}$ of the expgain and powerlaw files
$\bar{a}$	normalized gain a/a_o	unbarred a_w is the physical gain, 1/pN

Imported sources. Base chain (control law → boxed motion error) from `4state_del_hd.tex` / `5state_expgain_hd.tex`, imported in S3 with the $\bar{h} \to \bar{w}$ rename (map in S3 notes below); shape selection and Jacobian verification from `gainlaw_shape_selection.tex` and `aprime_jacobian_AB.tex`; truth curve from `two_sphere_truth_source.md`.

S1 notes — Gain law

Representation. The writing $\bar{a} = 1 - (1 + (\bar{w} - \bar{w}_s)/b)^{-p}$ was fixed by the user (2026-07-31). It is algebraically identical to the length writing $1 - [\, b/((\bar{w} - \bar{w}_s) + b)\,]^p$ of `gainlaw_shape_selection.tex`;

the equivalence, the three parameter Jacobians in this writing, and the wall-shift identity $\partial\bar{a}/\partial\bar{w}_s = -\bar{a}'$ were all checked symbolically (MATLAB, residual exactly 0, 2026-07-31), so every verified asset of `aprime_jacobian_AB.tex` carries over by rewriting alone.

Contact property and the amplitude-writing trap. At $\bar{w} = \bar{w}_s$ the base is 1, so $\bar{a} = 0$ for any (b, p): $\bar{w}_s$ is the contact position by construction, and the unknown offset $\bar{w}_s - 1$ is what the estimator will carry. The bare amplitude writing $1 - \lambda_{amp}(\bar{w} - \bar{w}_s)^{-p}$ loses this: fitted against the truth it returns $\bar{w}_s = -0.063$, which is not a surface position, and it drags fractional powers and ln terms into every Jacobian. Do not use it. (Audit note in `gainlaw_shape_selection.tex`.)

Limits and anchors (removed from the main file 2026-07-31, kept here). Near wall $\bar{a} = (p/b)(\bar{w} - \bar{w}_s) + O((\bar{w} - \bar{w}_s)^2)$; far field $1 - \bar{a} \to (b/(\bar{w} - \bar{w}_s))^p$. Both anchors are derived, not fitted (truth = two-sphere $\perp$, Jeffrey & Onishi 1984): the far-field truth $1 - \bar{a} \sim (9/8)/(\bar{w} - \bar{w}_s)$ forces $p = 1$ and $b = 9/8$ (the reflection coefficient); the near-wall truth slope is 1 while the model slope at the anchored pair is $p/b = 8/9$ — the truth sits $9/8 = 1.125$ above the model (+12.5% of the model; equivalently the model is 11.1% below the truth). The two ends cannot be satisfied simultaneously: this tension (quoted as 12.5%, direction as stated) is the quantified shape imperfection, and it funds the parameter prior widths in S10 (the same role the interpolation-gap sup plays for the expgain and powerlaw priors).

Fit record (minimax over $\bar{w} \in [1.1, 10]$, 2×10^4 samples): three parameters free give sup error 0.429% at

$(b,\ p,\ \bar{w}_s) = (1.0566,\ 0.9563,\ 0.9936)$;

$(p{=}1,\ b{=}9/8)$ pinned with $\bar{w}_s$ the only estimate: 0.779%. Form A under the same protocol (`gainlaw_shape_selection.tex`): 4.256%, about 10× worse; its two asymptotic anchors have no joint solution.

Why a_o is fixed, not a state. Two independent reasons. (i) *Honesty*: the y_2 variance-readout chain currently carries the open motion-time defect (+4 ∼ 5%, defect 2); a freed $\hat{a}_o$ acts as a bias sponge for it — measured on the expgain 7a prototype, $\hat{a}_o$ parks at +1.2% off truth while the overall RMS improves (0.477% → 0.144%). That RMS comparison is exactly the no-trade-off fingerprint: both numbers share the polluted readout chain, so it must not decide the question. Fixed until defect 2 is re-derived (`Cdpmr_Cn_derivation.tex`). (ii) *No absorber exists*: the far-field limit of $\bar{a}$ is pinned at 1 for every $(b, p, \bar{w}_s)$, so no shape parameter can mimic an amplitude error; an a_o error is an irreducible multiplicative bias, uniform in height ($\delta a_w/a_w = \delta a_o/a_o$ at every $\bar{w}$), bounded by the calibration tolerance chain $\delta a_o/a_o = \sqrt{(2\,\delta R/R)^2 + (\delta\eta/\eta)^2} \approx 2.8\%$ ($a_o = \Delta t/(\gamma_N R)$, $\gamma_N = 6\pi\eta R$).

Acceptance consequences (S11): inject a_o perturbations of ±2.8% and measure the leakage into $\hat{b}_w$, $\hat{p}_w$, $\hat{\bar{w}}_s$ (parameter-invariance style test); revisit the fixed-vs-free question only after defect 2 closes.

$p = 1$ **special case.** With the far-side anchors in force the law and all its Jacobians are rational (no `pow`/`exp`/`log`), which the 10 kHz FPGA loop can evaluate directly.

S2 notes — Gain rate and parameter Jacobians

Provenance and verification. All three Jacobians are the Form B set of `aprime_jacobian_AB.tex`, rewritten into the photo representation; each scalar bracket equals $\partial \ln \bar{a}'/\partial\theta$ (log-derivative factorization $J_\theta = [\text{scalar}] \cdot \bar{a}'$), and each was re-verified symbolically in this writing (MATLAB symbolic check, residual exactly 0, 2026-07-31). The acceptance protocol for any future re-expansion is Richardson ratio ≈ 100 with a mandatory negative control collapsing to ≈ 10, per coordinate direction (`aprime_jacobian_AB.tex`).

Menq-note comparison. The companion note's J_b, J_p were written for Form A and carry sign errors (the $-$ from $d\,e^{-u}/d\theta$ was dropped; its J_{w_s} is correct, which proves the three were mutually inconsistent). The form swap replaces that whole block; nothing needs patching line by line.

Expansion validity. First order in $(e_{b_w}, e_{p_w}, e_{\bar{w}_s})$; the $O(\|e\|^2)$ remainder is dropped — declared here once so the main file can stay clean. Both $\bar{a}'$ evaluations share the anchor point $\bar{w}_d - \hat{\bar{w}}_s$, so this expansion covers parameter errors only; the evaluation-point error (command vs true position, $O(\bar{a}''\delta\bar{w})$) is handled where the process model is built (S4), not here.

Identifiability warning (for S10, not a reason to change the form): all three J's multiply the same scalar excitation in the error dynamics, so per step the parameter subspace is excited along a single direction; measured on the 3-parameter fit: Fisher condition $\sim$270, $\cos(b, p\text{-directions}) = 0.979$, $\cos(b, w_s) = 0.642$. Handled by priors and trajectory design; individual $\hat{b}_w$, $\hat{p}_w$ must not be quoted separately (four-rules note of the 2026-07-29 memory).

S3 notes — True dynamics

Import and rename map. The boxed $\delta\bar{w}$ recursion and the F_{dw}, ε_w definitions are the boxed motion-error result of the base chain (`4state_del_hd.tex` / `5state_expgain_hd.tex` p.1–2), imported with the axis rename:

there	$\bar{h}$	$\delta\bar{h}$	$\Delta\bar{h}_d$	F_{dh}	ε_h	n_h	f_{dh}
here	$\bar{w}$	$\delta\bar{w}$	$\Delta\bar{w}_d$	F_{dw}	ε_w	n_w	f_{dw}

The rename is mechanical; the equations were not re-derived here, and their SSOT remains the base-chain file.

Imported premises (carried by the boxed recursion; re-audit list): (1) measurement delay exactly d steps, no jitter; (2) unmodeled disturbance $\delta\bar{w}_D \equiv 0$ (dropped from the control law and hereafter); (3) closed-loop pole λ_c known exactly to the estimator; (4) gain-error back-step: $a_w[k-i] - \hat{a}_w[k-i] \approx e_{a_w}[k]$ over the d-step window — the dropped term $(\bar{w}_d[k] - \bar{w}_d[k-i])(a'_w - \hat{a}'_w)$ is form-dependent and must be re-bounded with Form B's $\bar{a}'$, $\bar{a}''$ before S11 signs off (open TODO, tracked here).

Delay line. The states $\delta\bar{w}_1, \ldots, \delta\bar{w}_3$ follow the State-section convention of `5state_expgain_hd` (the measurement carries $\delta\bar{w}[k-d]$, i.e. $\delta\bar{w}_1$); the companion note has no delay-line states, which is why its error system is not closed — bringing the delay line into the true dynamics here is the closure fix. $d = 2$ is the project's canonical setup; the sums stay generic in d.

Reading of the true-gain line. Written at $[k{+}1]$ in the companion note's composition form $\bar{a}_w[k+1] = \bar{a}_w(\bar{w}[k+1])$: the true gain is evaluated at the true position $\bar{w} = \bar{w}_d - \delta\bar{w}_3$, i.e. it trails the command by the tracking error, and this $[k{+}1]$ form is what the process model (S4) Taylor-expands. The estimator (S5) will evaluate at $\bar{w}_d$ with the estimated parameters, and the difference splits into a parameter part (S2 Jacobians) and an evaluation-point part ($O(\bar{a}''\delta\bar{w}_3)$, bounded in S4). e_{a_w} appears inside the true $\delta\bar{w}_3$ recursion (self-modulation: the true trajectory depends on the estimation error through the control); this coupling is inherited from the base chain, not introduced here.

Constants. $\bar{w}_s$, b_w, p_w carry $[k+1] = [k]$ with zero process noise; the honesty condition that legitimizes $Q_\theta = 0$ ($\sup|\theta_{\text{eff}} - \theta_0| \lesssim \sqrt{P[0]}$) is stated and tested in S10/S11, not assumed silently.

S4 notes — Process model

Decision record (user, 2026-07-31). S4 follows the companion note's structure: $\bar{a}_w$ is a state propagated by $\bar{a}'$, and b_w, p_w are *not* pinned — the full parameter set rides along, so the closed system is 7 states: $[\delta\bar{w}_1, \delta\bar{w}_2, \delta\bar{w}_3, \bar{a}_w, b_w, p_w, \bar{w}_s]$ (parameter order $b_w, p_w, \bar{w}_s$ throughout, user 2026-07-31). Identifiability of $(b_w, p_w, \bar{w}_s)$ is handled by priors and trajectory design (S10); individual $\hat{b}_w$, $\hat{p}_w$ must not be quoted separately (S2 warning).

First-order Taylor and its two dropped terms. The boxed row is $\bar{a}_w[k+1] = \bar{a}_w(\bar{w}[k+1])$ expanded to first order about $\bar{w}[k]$, with $\bar{a}'$ evaluated at the desired gap (the note's highlighted line). Dropped without statement in the note, declared here: (i) the Taylor remainder $O(\bar{a}'' \cdot (\text{increment})^2)$; (ii) the evaluation-point error $O(\bar{a}'' \cdot \delta\bar{w}_3 \cdot \text{increment})$ from using $\bar{w}_d - \bar{w}_s$ instead of $\bar{w}[k] - \bar{w}_s$. Closed form

$$\bar{a}'' = -\frac{p(p+1)}{b^2}\left(1 + \frac{\bar{w} - \bar{w}_s}{b}\right)^{-p-2}, \qquad \sup_{\bar{w} > \bar{w}_s} |\bar{a}''| = \frac{p(p+1)}{b^2},$$

the sup attained at contact (nominal $(p, b) = (1, \frac{9}{8})$: $\frac{128}{81} \approx 1.58$), so both bounds are computable once the trajectory class is fixed (to be instantiated in S11).

Full-state assembly. The section closes with the state vector $\mathbf{x} = [\delta\bar{w}_1, \delta\bar{w}_2, \delta\bar{w}_3, \bar{a}_w, b_w, p_w, \bar{w}_s]^\top$ ($d = 2$) and the complete boxed true-state block — the house form of `5state_expgain_hd`'s "True state equation". Rows 1–3 repeat the S3 box (deliberate, same as the house doc); row 4 is the gain process model; rows 5–7 the constants. The standalone gain row above the block is the derivation step (Taylor of the S3 composition), left unboxed.

Structural risk record (defect 1). Differential propagation integrates a static curve as a time recursion. On the expgain law this produced the measured defect-1 signature: descent peak 11.03% (differential) vs 5.05% (algebraic evaluation); adding the y_2 measurement *worsens* $\times 2.05$ vs *improves* $\times 0.70$; constants' observability 159–7886$\times$ worse. The mechanism is structural and expected to transfer, but it has *not* been measured on Form B — hence the S11 fingerprint criterion is mandatory, and the algebraic alternative (evaluate $\hat{\bar{a}}_w$ from $(\hat{\theta}, \bar{w}_d)$ each step; identical Jacobian machinery, one state row differs) is kept on record here as the pivot plan.

Observability note. With $\bar{a}_w$ a state, the constants influence the filter only through the $\bar{a}$-row coupling $J_\theta[k] \cdot (\text{increment}) \cdot e_\theta$: during hold the increment ≈ 0 and $(b_w, p_w, \bar{w}_s)$ are unobservable (hold desert); motion is required. The exact channel $\partial\bar{a}/\partial\bar{w}_s = -\bar{a}'$ makes $e_{\bar{w}_s}$ and $e_{\bar{a}_w}$ instantaneously collinear in any gain readout; separation comes only from height variation. This drives the trajectory design of the S11 injection test.

Noise container preview (S7). The noise entering the $\bar{a}$-row is $\bar{a}' \cdot \varepsilon_w \to$ rank-one Q block across $(\delta\bar{w}_3, \bar{a}_w)$ (expgain pattern). The Taylor and evaluation-point truncations are coherent, trajectory-driven errors and must *not* be put into Q (container rule); they are exactly what the S11 fingerprint test watches.

S5 notes — Estimator

Mirror construction. The estimator is the process model with every quantity it cannot know removed and every remaining quantity replaced by its estimate: $F_{dw}\, e_{\bar{a}_w}$ is dropped because $e_{\bar{a}_w}$ is unknown (the estimator does know F_{dw} — it computes the control — but not the error it multiplies); ε_w is dropped as zero-mean noise. $\delta\hat{\bar{w}}_3$ therefore contracts with the ideal closed-loop pole λ_c alone. The gain row mirrors S4 with $\hat{\bar{a}}'[k]$ evaluated at the estimates (the companion note's second highlighted line): run-time computable from $\bar{w}_d$ and the state estimates only — c-free by construction.

Prediction-only, by design. Following the companion note, S5 carries no correction terms; the gains enter with the measurement (S7) and the filter equations (S9). (`5state_expgain_hd` folds the ℓ_{ij} update terms into its Estimator section instead — a layout difference only, noted to prevent a false mismatch reading.)

Closure fix. The companion note has no $\delta\hat{\bar{w}}_j$ states, so its $e_w = \delta\bar{w} - \delta\hat{\bar{w}}$ had no dynamics and the error system did not close. With the hat delay line present, $e_{\delta\bar{w}_j} = \delta\bar{w}_j - \delta\hat{\bar{w}}_j$ acquires the shift dynamics, and S6 (True − Estimator) closes the 7-dimensional error system.

S6 notes — Error dynamics

Derivation split (the section's single first-order declaration). The exact difference of the two gain rows decomposes as

$$(\bar{a}' - \hat{\bar{a}}') \cdot \big[\Delta\bar{w}_d + (1-\lambda_c)\delta\hat{\bar{w}}_3\big] \;+\; \bar{a}' \cdot \big[(1-\lambda_c)e_{\delta\bar{w}_3} + F_{dw}e_{\bar{a}_w} + \varepsilon_w\big].$$

The first factor is replaced by the S2 Jacobian expansion (evaluated at the estimates); in the second group $\bar{a}' \to \hat{\bar{a}}'$, which introduces only second-order terms $O\big((\bar{a}' - \hat{\bar{a}}') \cdot (e, \varepsilon_w)\big)$ (error-times-error and error-times-noise, the latter landing in q_4's coefficient) — dropped, declared here. The row-wise result is structurally identical to the companion note with its e_w read as our $e_{\delta\bar{w}_3}$.

Prediction-only form. The boxed error state equation carries no correction: $-\mathbf{L}[k]\mathbf{H}[k]\mathbf{e}[k]$ joins in S9 once the measurement (S7) exists. (`5state_expgain_hd` folds the ℓ_{ij} terms into its row-wise error dynamics instead — layout difference only.)

q structure and a known cross-correlation. $q_3 = -\varepsilon_w$, $q_4 = \hat{\bar{a}}'\varepsilon_w = -\hat{\bar{a}}'q_3$: the gain block is rank one, the expgain pattern. ε_w contains $(1-\lambda_c)n_w$, so $\mathbf{q}$ correlates with the y_1 measurement noise ($E[\mathbf{q}\,\mathbf{r}^\top] \neq 0$) — the same declared-open item as `5state_expgain_hd`'s NOTES; to be handled or bounded in S8.

Single-direction excitation, now visible. Columns 5–7 of $\mathbf{F}_e$'s row 4 share the common factor $[\Delta\bar{w}_d + (1-\lambda_c)\delta\hat{\bar{w}}_3]$: per step the three parameter errors are excited along the single direction spanned by $(J_{b_w}, J_{p_w}, J_{w_s})$ — the S2 identifiability warning made concrete. Hold $\Rightarrow$ the factor $\approx 0 \Rightarrow$ parameter hold-desert (S4 note); this drives the S10 priors and the S11 trajectory design.

S7 notes — Measurement and innovation

Readout provenance. $y_2 = \bar{a}_{wm} = a_{wm}/a_o$ is the variance-readout gain: the physical a_{wm} is reconstructed from $\hat{\sigma}^2_{\delta\bar{w}_r}$ through the C_{dpmr}/C_n chain, which arrives in S8 together with its OPEN flag (defect 2: the chain assumes a stationary, noise-only-driven loop; motion-time excess $+4 \sim 5\%$). The normalization by a_o is the second and last place a_o appears (Conventions).

Delay back-off — house form by decision (user, 2026-07-31). The back-off uses the command displacement only, $\bar{a}_w[k-d] = \bar{a}_w[k] - \bar{a}'\nabla_d\bar{w}_d$, matching `5state_expgain_hd` exactly. Declared truncations: (i) the tracking-error difference over the delay window is dropped — the exact increment is $\nabla_d\bar{w}_d - \delta\bar{w}_3 + \delta\bar{w}_1$, so the dropped term is $\hat{\bar{a}}'(\delta\bar{w}_3 - \delta\bar{w}_1)$, zero in hold and nonzero in motion transients (the house NOTES record this channel 1.3× low); (ii) the back-off Taylor is first order, remainder

$O(\bar{a}''\,\nabla_d\bar{w}_d^2)$; (iii) $\bar{a}'$ in the back-off is evaluated at the command point — the same evaluation-point class declared in S4, $O(\bar{a}''\,\delta\bar{w}_3\,\nabla_d\bar{w}_d)$. Consequence for S8: since $\mathbf{H}$ no longer carries the delay-window channel, R_2 must — the house doc's $+\,d\,Q_{44}$ term. The exact-$\mathbf{H}$ alternative ($\mathbf{H}$ row 2 columns $1/3 = \mp\hat{\bar{a}}'$) was derived and symbolically verified (constrained $\delta\bar{w}_3$-dependence cancels exactly; $\delta\bar{w}_1$-sensitivity matches the exact readout; 2026-07-31) and is kept on record here as an available upgrade — it is the 7b-analogue of 7a's $H(2,1) = -a'$ fix.

Form B removes a house-doc truncation. In `5state_expgain_hd`, $\partial y_2/\partial a_h = 1 + \nabla_d\bar{h}_d\hat{B} \to 1$ was itself a truncation ($\sim 0.4\%$). Here $\bar{a}'$ is built from $(b_w, p_w, \bar{w}_s)$ and does not depend on the state $\bar{a}_w$, so $\partial y_2/\partial\bar{a}_w = 1$ exactly — nothing is dropped.

Whitening pointer. y_2 as written is the raw EWMA readout; it is AR(1), and treating it as white overestimates its information 39× (measured; a_{cov}-invariance FAIL 3.14% raw vs PASS whitened). The whitening transform is introduced together with R_2 in S8, modifying $(y_2, \mathbf{H}\text{ row }2, R_2)$ as one unit — S8 discussion item.

First-order declaration. In e_{y_2} the mixed second-order terms $O\big((\bar{a}' - \hat{\bar{a}}')\cdot e_{\delta\bar{w}}\big)$ are dropped, same one-declaration convention as S6.

S8 notes — Process and measurement noise

Declared exclusions (house-doc lineage; item (i) REVISED 2026-08-01). (i) ORIGINAL: the past thermal steps $(1-\lambda_c)\sum\bar{a}_w[k-i]\bar{f}_T[k-i]$ and the n_w feedthrough stayed out of Q (house-doc declaration). REVISED to the validated 3-component container (the D3 precedent of `reference/eq17_analysis/kf_canonical_spec.md` §5, Vpersonal lineage, empirically validated emp/closed ≈ 1.0 on the 6-state): $Q_{33} = \mathrm{Var}(\varepsilon_w)$ in full — current thermal step, thermal history at $(1-\lambda_c)^2$, and $(1-\lambda_c)^2\sigma_{n_w}^2$ feedthrough — with only the *cross-step correlations* of ε_w dropped (independence approximation; ε_w is MA(d) across steps, declared). The 6-state's third component (randgain) does not apply to Form B: the gain-error-times-force channel lives coherently in $F_{dw}e_{\bar{a}_w}$ via the back-step, and the shape parameters carry no random walk. Motivation for the revision: the 2026-08-01 audit traced the descent-phase low-frequency leakage into the free parameters partly to the under-modelled (colored) ε_w; the rank-one gain block ($q_4 = -\hat{\bar{a}}'q_3$) is unaffected by the container's internal composition. (ii) ε_w contains $-(1-\lambda_c)\,n_w[k-d]$ (sign and lag corrected 2026-08-01 by log forensics, regression tap -0.296 at $k-d$; the earlier $+(1-\lambda_c)n_w[k]$ was wrong — the variance term $(1-\lambda_c)^2\sigma_{n_w}^2$ is unchanged), so $E[q_3[k]\,r_1[k-d]] = +(1-\lambda_c)\sigma_{n_w}^2 \neq 0$ is dropped (the standard-KF independence assumption is kept; same treatment as the D3 precedent; scoped out of the 2026-08-01 revision by user decision; order $\sigma_{n_w}^2/\sigma_T^2 \approx 9\%$ times $(1-\lambda_c)$, subdominant). (iii) Run time evaluates every Q entry at the current estimate — stated once here, fixing the evaluation-point asymmetry the house doc left undeclared. (iv) $d\,Q_{44}$ in R_2 carries the delay-window gain wander because S7's house back-off leaves that channel out of $\mathbf{H}$ (S7 notes; the house NOTES record this channel 1.3× low).

Defect 2: RETRACTED (2026-08-01). C_{dpmr}, C_n are derived (`Cdpmr_Cn_derivation.tex`) for a *stationary, noise-only-driven* closed loop, and the 2026-07-28 case held that commanded motion adds a $+4 \sim 5\%$ readout excess. Re-adjudicated on the timing-fixed machinery: at N=48 seeds the oscillation-window excess is +1.33% (95% CI $[-0.02, +2.67]\%$, contains zero) and the paired hold−osc difference is $-0.44\%\pm0.90\%$ — the motion-time framing dissolves; the old +4.4% was a 6-seed sampling artifact (the house seed set is a 1-in-18 high draw; seed 11 is the population maximum). Mechanism executions on record: the seven 07-28 falsifications stand; the gain-modulation candidate was killed by the paired far-field test (modulation power ×7219 down, bias moved $-0.11\%\pm0.11\%$); the EWMA-lag candidate was killed quantitatively (25× power short, closure with the falsification ledger). The *live* open item is the per-seed finite-window readout fluctuation (std 5–6%, fixed per noise realization, independent of height and motion) — formerly billed as the c2/c3 ratchet fuel, REFUTED by the 2026-08-01 y2_off ablation: the ratchet is y_1-fed (ε_w MA(2), the mainline work item; the shelved β_w container for the y_2 DC is archived in `archive/formB_beta_w_shelved_2026-08-01.tex`); a possible $\sim +1.2\%$ stationary offset remains unresolved (needs $N \sim 200$). Readout-chain numbers from 6 seeds carry SEM $\approx 2.6\%$ and must not be quoted at the 1% level.

MA(2) augmentation — verification and ablation record (2026-08-01). The main-file augmentation block is the mainline fix for the c2/c3 seed ratchet. Verified by a three-agent audit (independent blind re-derivation + log forensics + adversarial judge, overall CONFIRMED): recursion-residual reconstruction of ε_w from the simulation logs closes at corr 0.99988 (residual 1.5%, hold window); empirical lag ratios $C(1)/C(0) = 0.325$, $C(2)/C(0) = 0.245$, $C(3+) \leq 0.3\%$, against predicted 0.33/0.24/0; empirical DC sum 2.149 vs predicted 2.162 (99.4%). The y_1 innovation carries the same signature ($\rho_{1..5} = [0.299, 0.243, \sim 0, \sim 0, \sim 0]$, hold window, 6 seeds) — the acceptance is whitening those two lags. Three-arm ablation, C2 config (lock p, estimate b, $\bar{w}_s$), N=20 paired seeds: (a) white container — b budget aggregate 1.963 FAIL (≤ 1.520), desc $5.68 \pm 3.97\%$; (b) DC-matched white (diagnostic knob `q33_dc_match`, thermal $Q_{33} \times 2.1695$) — b budget 1.074 PASS, desc $3.85 \pm 2.45\%$, but the $\hat{\bar{w}}_s$ tail persists (seed 11 -20%); (c) exact augmentation — see the implementation record. Implementation cautions: the off-diagonal Q_{38}, Q_{48} couplings are load-bearing (diagonal-only Q silently wrong); the memory-state predict must respect the buffered $[k-1]$ timing convention of the 2026-08-01 timing fix; the m rows inherit the per-step $\bar{a}_w[k]$ time-varying variance (constant-σ_T^2 only in the stationary windows). Residual out of scope: the osc-window closure leaves a 15% trajectory-coupled residual (loop-gain modulation outside the S3 linearization) — the augmentation is exact for the hold spectrum.

y_2 self-echo — discovery, derivation, fix (2026-08-01). The suspected "frozen" per-seed y_2 bias ($\sigma_c = 6.1\%$, $2\times$ the flat-χ^2 closed form) was RETRACTED on fresh seeds: N=40 new seeds under an oracle loop give cross-window corr 0.022, window stds at 99%/87% of the χ^2 predictions, and zero mean offset (the +1.2% stationary-offset suspicion dies with it) — the 6.1% was a seeds-1:20 sampling artifact, the third seed-set trap of the record. The REAL defect: the reading tracks the *applied* control gain with sensitivity S (loop-pole shift). Verified three ways: (i) exact 6-state Lyapunov model of the mismatched loop (states $x, x[k-1], x[k-2], u[k-1], u[k-2]$, a_{pd} tracker) matches the logged $\mathrm{Var}(\delta\bar{w}_r)$ to 0.3% ($4k_BT$ thermal convention of `calc_thermal_force`) and gives $S = +0.319$ far field; (ii) paired-forcing measurement (same seed, $a_{ctrl} = a_{ff}(1 \pm 0.10)$, χ^2 draws cancel in the pair) gives $S = +0.323 \pm 0.043$; (iii) $S(\bar{w})$ is near-constant (0.310–0.319). Fix (`y2_echo_corr`, production default; legacy arm bit-identical when false): $\mathbf{H}_2 \times (1 - S)$ with the per-step thermal/noise blend. Effects at N=20: $\hat{\bar{w}}_s$ honesty 1.12 → 0.93, RMS 0.070 → 0.065, b budget 1.088 → 1.053, hold-window y_2-door push ×0.82.

Efficiency audit — the filter sits on the information bound (2026-08-02). Honesty (spread = claimed $\sqrt{P}$) does not imply efficiency, so the CRLB was probed by paired sensitivity (true wall ±0.005, same noise realization): measurement-noise-only bound $\sigma \geq 0.006$ (unreachable, thermal dominates the record), per-sample-independent bound 0.037–0.039, and correlation-corrected ($\sum \rho \approx 5.7$ at $\lambda_c = 0.7$) ≈ 0.09; the filter's no-prior single-run equivalent is 0.080 — *at* the bound; no estimator class recovers meaningfully more from one 4.8 s record. Information budget: descent 32%, osc 35% (≈ 27 units/s near wall), state-memory residue 24%; a no-motion ablation (hold-only, ±0.05 injection) freezes $\hat{\bar{w}}_s$ at $1 \pm 0.1\%$ and cannot see the injected wall — all $\bar{w}_s$ information comes from wall-ward motion.

Root cure — $\bar{w}_s$ as a calibration constant (doc page c5). Per-run scatter is the honest posterior realization; the cure is the information ledger: sequential posterior handoff (run n's $(\hat{\bar{w}}_s, \sqrt{P_{w_s}}) =$

run $n+1$'s prior). Six chains $\times$ 8 runs, fresh noise each run, true wall 1.05: cross-chain spread $0.089 \to 0.020$, tracking the claimed $\sqrt{P}$ throughout; operational runs on the calibrated prior: $\hat{\bar{w}}_s$ error std 1.3%, descent peak 1.69% $\approx$ the config-1 locked floor. Next lever (measured, not yet built): the wall is SHARED across axes — per-axis errors are independent with comparable spread, so a shared $\hat{\bar{w}}_s$ buys $\div\sqrt{3}$; prerequisite: the parallel-axis gain law currently reuses the perpendicular anchors ($b = 9/8$ vs the parallel truth 9/16), which biases the x/y wall estimates to ~ 0.6 — the stage-1 derivation of the development plan.

Parallel-axis law — Stage 1a record (2026-08-02). The x/y filters ran the perpendicular law ($b = 9/8$, $p = 1$) against the parallel truth: $c_{\|}$ leaves contact with Goldman log behavior, which no Form B member reproduces — the best representation slides its origin INSIDE the wall ("the parallel axis cannot report the wall", independent finding of the shape-selection line, `gainlaw_para_origin_shift.m`). Consequences before the fix: x/y gain estimation carried desc 34.8/34.1%, osc RMS 21.0/20.5%, hold $-25.0/-22.6\%$ (6-seed means), and their $\hat{\bar{w}}_s$ faithfully converged to the law's effective origin (≈ 0.6), poisoning any cross-axis fusion. Fix (S1 parallel block; `par_law`, driver default true, `_nopar` legacy arm): the driver derives $(b_{\|}, p_{\|}, \bar{w}_{s0,\|})$ at init by the 3-parameter minimax fit of the law to the parallel GAIN $1/c_{\|}$ (fitting the deficit instead diverges — labelling trap) on the same planned envelope as the (b, p) priors, plus boundary widths and the representation floor $P_{\bar{a}\bar{a},\|}[0]$; the x/y slots 5–7 are locked at the constants and their law origin rides the z filter's $\hat{\bar{w}}_s$ one-way ($+\Delta_{\|}$). Acceptance (all PASS): legacy arm bit-identical (3 axes); z bit-identical with the law on; x/y health desc $\to 0.92/0.86\%$, hold $\to -0.04/+0.01\%$ ($\sim 40\times$, doc page c6); under a 1.05 injection the x/y residual (+0.8% hold) is the pushforward of z's own $\hat{\bar{w}}_s$ error through $\partial\bar{a}/\partial\bar{w}_s$ — the intended one-way coupling, not a defect. The x/y $\bar{w}_s$-information question (fusion, Stage 1b/2) is decided by experiment, with the derived systematic widths as the honest cap.

Stage 1b verdict — $\bar{w}_s$ stays z-only (2026-08-02). Duel instrument `par_ws_free` (x/y keep the parallel constants but estimate their OWN physical $\bar{w}_s$; regression bit-identical when off): over the canonical run the x/y claimed $\sqrt{P_{w_s}}$ contracts only $0.111 \to 0.106$, i.e. $I_x/I_z = 0.061$, $I_y/I_z = 0.071$ — on the parallel axes y_1 carries no commanded-force regressor, so their only $\bar{w}_s$ channel is the weak (whitened, echo-discounted) y_2 column. Power analysis kills the planned chain duel: the predicted fusion gain is $\sqrt{1 + I_x/I_z + I_y/I_z} - 1 \approx 6\%$ in σ, below a 6-chain experiment's resolution (SEM $\sim 30\%$) and priced against importing the SHARED $\Delta_{\|}$ systematic (± 0.015, common to x and y). Verdict: $\bar{w}_s$ estimation stays z-only; the x/y law origins keep the one-way z feed; the shared-state fusion stage is dropped from the plan; `par_ws_free` is kept as the reopening instrument (doc page c7).

The correct (whitened) y_2 treatment — production default. $\hat{\sigma}^2$ is an EWMA, so y_2 is AR(1) with pole $(1 - a_{cov})$: K_{var} fixes the per-sample *variance* but not the sample-to-sample *correlation.* Fed raw at every step, the filter overcounts information by $(2 - a_{cov})/a_{cov} \approx 39\times$ (exact analytic factor; the 2026-07-27 incident — the fake 0.92–0.95 results). Whitening feeds the AR(1) innovation instead,

$$\tilde{y}_2[k] = y_2[k] - (1 - a_{cov})\, y_2[k-1] = a_{cov}\, s[k],$$

i.e. the instantaneous readout $s[k]$ with per-sample variance $2\, IF_{var}(\bar{a}_w + \xi)^2$ $(+\, d\, Q_{44})$ and (approximately) independent samples — the same information rate, in an honest container. Verified: a_{cov}-invariance scan raw FAIL 3.14% ($P[0]$ budget blown 4.645×) vs whitened PASS 0.00–0.02%; whitening is the production behaviour in both controllers — unconditional in `motion_control_law_5state_powerlaw.m`, `y2_whiten` toggle (default true) in `_5state_expgain.m` — the implementation SSOT. The main text keeps the raw, white-treated form by decision (user, 2026-07-31) to mirror the companion documents.

S9 notes — Filter equations (not in the main file)

Both companion documents stop before the filter proper; by decision (user, 2026-07-31) the filter equations live here.

$$
\begin{aligned}
&\text{predict:}\quad \hat{\mathbf{x}}^-[k+1] = \text{S5 rows}, \qquad \mathbf{P}^-[k+1] = \mathbf{F}_e[k]\,\mathbf{P}[k]\,\mathbf{F}_e^\top[k] + \mathbf{Q}[k]\\
&\text{update:}\quad \mathbf{S}[k] = \mathbf{H}[k]\mathbf{P}^-[k]\mathbf{H}^\top[k] + \mathbf{R}[k], \qquad \mathbf{L}[k] = \mathbf{P}^-[k]\mathbf{H}^\top[k]\,\mathbf{S}^{-1}[k]\\
&\qquad\qquad \hat{\mathbf{x}}[k] = \hat{\mathbf{x}}^-[k] + \mathbf{L}[k]\,\mathbf{e}_y[k], \qquad \mathbf{P}[k] = (\mathbf{I}-\mathbf{LH})\,\mathbf{P}^-\,(\mathbf{I}-\mathbf{LH})^\top + \mathbf{LRL}^\top \quad \text{(Joseph)}
\end{aligned}
$$

This supplies the correction left open since S6: $\mathbf{e}[k+1] = \mathbf{F}_e\mathbf{e} - \mathbf{LHe} + \mathbf{q}$ (the house doc's boxed error state equation). With diagonal $\mathbf{R}$ the two channels may be applied as sequential scalar updates (code practice). $\mathbf{P}[0]$ from S10; DARE-based initialization per the implementation SSOT (`motion_control_law_5state_expgain_alg.m` precedent).

S10 notes — Seeds and priors (not in the main file)

Instantiated 2026-07-31 via the $P[0]$ six-step chain of `reference/shared/param_prior_rules.md`; the (b_w, p_w) widths below are derivation, not tuning. Numeric tool: the Form B branch of `verify_shape_exponent_bound.m` (`test_script/integration/`).

Seeds. $\hat{b}_w[0] = 9/8$, $\hat{p}_w[0] = 1$ (the two derived anchors), $\hat{\bar{w}}_s[0] = 1$ (nominal contact); $\hat{\bar{a}}_w[0] = \bar{a}(\bar{w}_d[0] - \hat{\bar{w}}_s[0], \hat{b}_w[0], \hat{p}_w[0])$; $\delta\hat{\bar{w}}_j[0]$ from the first measurements (or 0).

Six-step chain, instantiated for (b_w, p_w)**.** (1) $P[0] := E[(\text{true} - \text{seed})^2]$ (KF definition); (2) the seeds are the analytic anchors, so the ignorance source of the error is zero; (3) shape parameters have no true value — the defensible substitute is $\theta_{\text{eff}}(\bar{w})$, the value the truth curve demands at each height; (4) $Q_\theta = 0$ means one committed number serves the whole run, so the error at step k is $|\theta_0 - \theta_{\text{eff}}(\bar{w}[k])|$; (5) at $k = 0$ the height is unknown and the trajectory sweeps the domain, so the honest bound is the global sup; (6) $\sqrt{P[0]} = \sup|\theta_{\text{eff}} - \theta_0|$.

θ_{eff} **constructions — the reading choice.** Form B's deficit is $1 - \bar{a} = (1 + (\bar{w} - \bar{w}_s)/b)^{-p}$; the truth deficit is $(c-1)/c$. With $\bar{w}_s$ at its truth ($= 1$, the Tier-1 setup) the deficit still carries *two* parameters, so each needs its own reading, the other pinned at its anchor ($g := \bar{w} - 1$, the gap):

- p_{eff} by deficit log-log *slope*, $b = 9/8$ pinned. An exponent is a slope in log-log coordinates — the house precedent: the expgain and powerlaw θ_{eff} readings are both slope readings. Against the shape coordinate $1 + g/b$:

  $$p_{\text{eff}}(\bar{w}) = -\big(g + \tfrac{9}{8}\big)\,\frac{c'}{c\,(c-1)}.$$

- b_{eff} by deficit *level*, $p = 1$ pinned. b is a length, not an exponent, and the slope reading is degenerate for it: far field the log-log slope tends to p for *every* b — it reads p's channel and misses even b's own far anchor, so it cannot price the b commitment. The level reading inverts the pinned-p deficit $b/(g+b)$ for b:

  $$b_{\text{eff}}(\bar{w}) = (c-1)\,(\bar{w} - 1).$$

Endpoints and numbers (plant truth $c_\perp$). $b_{\text{eff}} : 1 \to 9/8$ and $p_{\text{eff}} : 9/8 \to 1$ from contact to far field: each reading travels exactly the 12.5% anchor tension of S1 (near-wall truth slope 1 vs anchored model slope $p/b = 8/9$), surfacing at the contact end.

reading	θ_0	sup all	sup $[1.1, 10]$	sup $\bar{w} \geq 2$
b_{eff} (level)	9/8	0.1244	0.0798	0.0157
p_{eff} (slope)	1	0.1235	0.0373	0.0242

Both global sups sit at the contact end and are the anchor tension itself (0.1244/0.1235 differ from 1/8 only through the sweep starting at $\bar{w} = 1.001$).

Decision (user-approved 2026-07-31, SUPERSEDED by the revision below):

$$\sqrt{P_{bb}[0]} = \sqrt{P_{pp}[0]} = \tfrac{1}{8} = 0.125$$

— the global sup, which is the anchor tension in closed form (9/8 − 1); no truncation to a working range, per chain step (5).

Revision — envelope domain (user-approved 2026-08-01). Chain step (5)'s premise ("height unknown, sweeps the domain") is false for a *commanded* trajectory: the planned envelope is known before $k = 0$. Amended step (5): for a commanded trajectory the sup domain is the planned envelope, widened by a derived tracking margin; the global sup remains the fallback when the envelope is unknown. The Tier-1 audit measured the consequence of the global width on this scenario: 5–8× over-wide, giving the collinear ridge room to absorb low-frequency innovation content (rule 3's "too loose gets dragged by noise", observed as the seed-11 ratchet). Envelope for the canonical scenario: commanded trough 2.0 minus margin 0.1 (measured true-trough undershoot 0.033 plus measurement noise ~ 0.05, rounded up) to start height plus 1:

$$\sqrt{P_{bb}[0]} = 0.0157, \qquad \sqrt{P_{pp}[0]} = 0.0245 \qquad \text{(envelope } [1.9,\ 22.3]).$$

Boundary sensitivity (floor 1.90/1.95/2.00): b 0.0157 in all three — its sup sits at the interior point $\bar{w} \approx 4.3$, boundary-immune; p 0.0245/0.0244/0.0242 — its sup sits at the floor, 0.0003 spread. Ceiling 10/22.3/25: no change in either. The driver derives the triple at run time from its own scenario config (zero tuning: envelope = config fact, sup = offline truth-curve evaluation); the controller defaults keep the global-sup fallback.

Tier-1 shape floor. With $\bar{w}_s$ locked at 1, the (b, p) minimax fit over $\bar{w} \in [1.1, 10]$ against the plant truth reaches sup gain error 0.353% at $(b, p) = (1.0354,\ 0.9500)$; the anchored seed $(9/8,\ 1)$ gives sup 0.66%. The anchored-seed number is the shape floor used below: the gain error the seed pair carries beyond what parameter offsets explain — derived, not a tunable. Envelope revision (2026-08-01, domain-consistency fix for the audit's mixed-domain note): on the canonical envelope $[1.9, 22.3]$ the anchored-seed sup is 0.284% at the interior point $\bar{w} \approx 3.47$ (boundary-immune; identical for floors 1.90/1.95/2.00) — the floor now lives on the same domain as the P_{bb}/P_{pp} widths; 0.66% remains the $[1.1, 10]$ fallback record.

$P[0]$ **cross-blocks at init.** The gain seed is a deterministic function of the parameter seeds, so its error is the pushforward of theirs plus the shape floor:

$$P_{\bar{a}\theta}[0] = \frac{\partial \bar{a}}{\partial \theta} P_{\theta\theta}[0], \qquad P_{\bar{a}\bar{a}}[0] = \frac{\partial \bar{a}}{\partial \theta} P_{\theta\theta}[0] \frac{\partial \bar{a}}{\partial \theta}^{\top} + (\text{shape floor})^2,$$

with the shape floor on the same domain as $P_{\theta\theta}[0]$ (envelope: 0.00284; global-fallback: 0.0066), with the *level* partials $\partial\bar{a}/\partial\theta = [\partial\bar{a}/\partial b,\ \partial\bar{a}/\partial p,\ \partial\bar{a}/\partial\bar{w}_s]$ (note $\partial\bar{a}/\partial\bar{w}_s = -\bar{a}'$ exactly; these are NOT the S2 gain-rate Jacobians $J_\theta = \partial\bar{a}'/\partial\theta$ — the seed is computed by the law's level, so its error pushes forward through the level) evaluated at the start height and the seeds.

Remaining widths. $\bar{w}_s - 1$ width from approach/positioning calibration and the S11 injection protocol (Tier-1 locks $\hat{\bar{w}}_s$ at 1). a_o is not a state — its 2.8% tolerance is exercised by the S11 leakage test, not carried in $\mathbf{P}$.

Honesty and reporting. $Q_\theta = 0$ honesty condition: $\sup|\theta_{\text{eff}} - \theta_0| \lesssim \sqrt{P[0]}$ (charter constraint 4) — met with equality by construction on the chosen domain (after the envelope revision: on the planned envelope): the prior *is* the sup, i.e. TIGHT (1.00×) by design. Collinearity per the four rules: report $\hat{\bar{a}}$ only; never quote $\hat{b}_w$, $\hat{p}_w$ separately; Joseph/UD update; near-wall separate fit.

S11 notes — Acceptance criteria (not in the main file)

Stated before implementation (derivation-workflow rule 3):

1. Defect-1 fingerprint (S4 risk): adding y_2 must *improve* $\hat{\bar{a}}$ tracking; benchmark the descent peak against the algebraic-evaluation oracle.
2. a_{cov} invariance: scan 10×; whitened must PASS (raw is expected to FAIL — that failure is the fingerprint of the raw container, not of the filter).
3. $P[0]$ budget: $E[(\hat{x}_\infty - \hat{x}_0)^2] \leq P[0] - P[\infty]$.
   Calibrated reading (2026-08-01; the earlier per-seed hard ≤ 1 was statistically miscalibrated — for an honest filter the per-seed traversal ratio is $\sim \chi^2(1)$, so one of six seeds exceeds 1.89 with probability ≈ 0.67): acceptance = the 6-seed *mean* ratio $\leq 1 + 1.645\sqrt{2/6} = 1.949$ (one-sided 95% under the saturated bound $E[\text{ratio}] \leq 1$); per-seed ratios are printed as diagnostics, flagged only above $\chi^2_1(0.99) = 6.635$.
4. Shape bound: $\sup|\theta_{\text{eff}} - \theta_0| \lesssim \sqrt{P[0]}$ with the Form B θ_{eff}.
   Tool: adapted `verify_shape_exponent_bound.m`.
5. $\bar{w}_s$ injection: inject a known wall offset; the filter must converge to it, on a *moving* trajectory (hold desert, S4/S6 notes).
6. a_o leakage: perturb a_o by ±2.8%; measure the pull on $\hat{b}_w$, $\hat{p}_w$, $\hat{\bar{w}}_s$.
7. Truncation bounds: instantiate the S4/S7 $O(\bar{a}'')$ bounds ($\sup|\bar{a}''| = p(p+1)/b^2$, at contact) for the canonical trajectory class and confirm they are subdominant to the shape bound (item 4).